

Prob 1. (40 points) Assume we have a string, for which we need to apply a swap operation to encrypt it. The algorithm is to swap the last 2 chars for every three chars and return the number of pairs swapped. For example:

- For string `h`, there is not enough chars to swap, and the result should still be `h`. Return is 0.
- For string `he`, there is not enough chars to swap, and the result should still be `he`. Return is 0.
- For string `hel`, we have one pair chars to swap, and the result will be swapped to `hle`. Return is 1.
- For string `hello`, we still have one pair of chars to swap, and the result will be swapped to `hlelo`. Return is 1.
- For string `helloo`, we still have one pair of chars to swap, and the result will be swapped to `hlelo`. Return is 1.
- For string `hellow`, we will have two pairs of chars to swap, and the result will be swapped to `hlelwo`. Return is 2.

1. (20 pts) Write a C function `int mp_swap_last_2_chars_in_3_in_str_c(char *str)` to perform the swap without calling existing function.
2. (20 pts) Translate the above C function an assembly with a prototype being

```
int mp_swap_last_2_chars_in_3_in_str_s(char *str).
```

Prob 2. (60 pts) Assume we have a string of lower-case letters. Assume also we are given the following C function which is used to copy this string to a new string with all upper-case ones.

```
1 void mp_str_cpy_2_caps_while_c(char *dst, char *src) {
2     char ch = *src++;
3     while (ch) {
4         ch -= 0x20; // 'a' = 0x61, 'A' = 0x41
5         *dst++ = ch;
6         ch = *src++;
7     }
8     *dst++ = ch;
9 }
```

1. (30 pts) Translate the above C function to an assembly with a prototype being `mp_str_cpy_2_caps_while_s(char *dst, char *src)`.
2. (10 pts) Reimplement the above C function using a `while` loop with `break` to eliminate the repetition code with a prototype being `mp_str_cpy_2_caps_while_brk_c(char *dst, char *src)`.
3. (20 pts) Translate the `while` with `break` C function to an assembly with a prototype being `mp_str_cpy_2_caps_while_brk_s(char *dst, char *src)`.